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ORIGINAL ARTICLE

Athletes have more susceptibility to oxidative stress: Truth or myth? A study in swimmers

Susceptibilité accrue des athlètes au stress oxydant : réalité ou mythe? Étude chez des nageurs

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KEYWORDS
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Summary
Aims. — Exercise is associated with accelerated reactive oxygen species (ROS) generation and stimulation of the antioxidant defences (AD). In high intensity exercise ROS production may overwhelm the AD and induce adverse effects on health and performance. We aim to investigate the redox balance of adult-competitive swimmers against recreational exercise practitioners.
Methods. — Fifteen high-competition male swimmers (S) and 16 active men (AM), all aged between 18—25 years, participated in the study. Venous blood was collected at rest in the morning to assess oxidative and muscle damage markers, antioxidant enzymatic activities in erythrocytes, and non-enzymatic antioxidants in plasma. All parameters were assessed by photometry with the exception of thiobarbituric acid reactive substances (TBARS), assessed by fluorimetry. Bivariate correlations were performed to investigate the association between all redox balance markers. Partial correlation analysis was conducted to control for the exercise level.
Results. — Differences between groups were observed in red blood cells peroxidation (RBCPx—S: 39.2 ± 4.8; AM: $46.3 \pm 9.5\%$) and in meta-hemoglobin reductase (RmHb — S: 7.5 ± 2.1; AM: 9.1 ± 2.3 μmol/min.gHb), with S presenting lower mean values ($P < 0.05$). Controlling for the

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exercise level, adrenaline oxidase (AdOx) was inversely correlated with carotenoids (R = −0.596; *P* = 0.001) and directly with creatine kinase (CK) (R = 0.413; *P* = 0.045) and with TBARS (R = 0.532; *P* = 0.005); RBCPx was inversely correlated with acid phosphatase (ACP1) (R = −0.392; *P* = 0.048) and directly with glutathione reductase (GR) (R = 0.524; *P* = 0.006); GR was inversely correlated with TBARS (R = −0.404; *P* = 0.041); VitE was directly correlated with Carotenoids (R = 0.723; *P* = 0.001) and inversely with TBARS (R = 0.543; *P* = 0.004). Therefore: AD are effective at controlling ROS; VitE and carotenoids are important protective lipid antioxidants; higher TBARS may be associated to the stimulation of GR activity; muscle leakages associated to the AdOx activity which may result from ion leakage, intravascular hemolysis or inflammation.
Conclusion. — Our results suggest that competitive swimmers are not at higher risk of oxidative damage than recreational exercise practitioners.
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MOTS CLÉS
Antioxydants ;
Espèces réactives de l'oxygène ;
Athlètes ;
Aptitude aérobie

Résumé
Objectifs. — L'exercice est associé à la génération accélérée d'espèces réactives de l'oxygène (ERO) et à la stimulation des défenses antioxydantes (DA). Dans les exercices de haute intensité, la production d'ERO peut outrepasser les DA et induire des effets néfastes sur la santé et la performance. Nous avons voulu étudier l'équilibre redox de nageurs compétiteurs adultes comparés à des sportifs amateurs.
Matériels et méthodes. — Quinze nageurs de haut niveau (N) et 16 hommes physiquement actifs (HA), tous âgés de 18 à 25 ans ont participé à l'étude. Le sang veineux a été recueilli au repos le matin pour évaluer les marqueurs lésionnels oxydatifs et musculaires, les activités enzymatiques antioxydantes dans les érythrocytes et les antioxydants non enzymatiques dans le plasma. Tous les paramètres ont été évalués par photométrie, à l'exception des substances réactives à l'acide thiobarbiturique (TBARs), évaluées par fluorimétrie. Des corrélations bivariées ont été effectuées pour étudier l'association entre tous les marqueurs d'équilibre redox. Une analyse de corrélation partielle a été effectuée pour neutraliser l'effet du niveau d'exercice.
Résultats. — Des différences entre groupes ont été observées pour la susceptibilité à la peroxydation des globules rouges (GRPx — N : 39,2 ± 4,8 ; HA : 46,3 ± 9,5 %) et pour la méthémoglobine réductase (RmHb — N : 7,5 ± 2,1 ; HA : 9,1 ± 2,3 μmol/min.gHb), N représentant les nageurs qui ont des valeurs moyennes plus faibles ($p < 0,05$). En contrôlant pour le niveau d'exercice, l'adrénaline oxydase (AdOx) était inversement corrélée avec les caroténoïdes (R = −0,596 ; $p = 0,001$) et directement avec la créatine kinase (CK) (R = 0,413 ; $p = 0,045$) et avec les TBARs (R = 0,532 ; $p = 0,005$); GRPx était inversement corrélée avec la phosphatase acide (ACP1) (R = −0,392; $p = 0,048$) et directement avec la glutathion réductase (GR) (R = 0.524; $p = 0,006$); GR était inversement corrélé avec les TBARs (R = −0,404 ; $p = 0,041$); La VitE était positivement corrélée avec les caroténoïdes (R = 0,723 ; $p = 0,001$) et négativement avec les TBARs (R = 0,543 ; $p = 0,004$). Par conséquent : les DA sont efficaces pour contrôler les ERO ; la VitE et les caroténoïdes sont des antioxydants lipidiques protecteurs importants ; des TBARs plus élevés peuvent être associés à la stimulation de l'activité GR ; la fuite musculaire est associée à l'activité d'AdOx qui peut résulter d'une fuite d'ions, d'une hémolyse intravasculaire ou d'une inflammation.
Conclusion. — Nos résultats suggèrent que les nageurs de compétition ne sont pas plus à risque d'effets délétères du stress oxydant que les pratiquants d'exercices récréatifs.
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1. Introduction

Swimming is considered an endurance and predominantly aerobic sport nevertheless both aerobic and anaerobic systems are activated and swimmers undergo intense training throughout the season [1]. Skeletal muscle is a highly specialized tissue with excellent plasticity in response to external stimuli such as exercise and training. The repetitive muscle contractions during endurance training lead to a variety of physiological responses, including the increase in reactive oxygen species (ROS) production [2]. Following the discovery that muscular exercise promotes oxidative stress in humans, many investigators have demonstrated that prolonged endurance exercise or short-duration/high intensity exercise results in the increase of oxidative stress markers in muscle and in blood, therefore increasing lipid peroxidation

and protein oxidation [2—4]. The changes observed in oxidative stress markers measured concurrently in blood and in tissues have been similar in animal studies after several interventions [5] suggesting that blood can be used as a less invasive approach to assess oxidative balance in tissues.

In blood, erythrocytes are particularly susceptible to oxidative damage because of the high polyunsaturated fatty acid content of their membranes and the high cellular concentrations of hemoglobin and oxygen. These cells are equipped with an elaborate antioxidant defence system (AD) to which contribute several antioxidant enzymes [6].

Professional training with repeated bouts of high volume and intensity of exercise, may also lead to an increase in inflammation and a decrease in performance [3,7—9]. The strong increase of ROS due to the intense exercise can lead to a variety of adaptations in cells, depending upon location of ROS generation, their concentration, duration of ROS exposure, and individual training status, and act as important physiological regulatory signals though mechanisms such as post-translational redox modifications of cysteine thiols on proteins. Such signalling can regulate diverse biological functions such as the maintenance of tissue homeostasis, cell proliferation and differentiation, cell migration and regulation of transcriptional activity [10].

Thus ROS can have both beneficial and detrimental effects [7,11], where a single bout of exhaustive exercise has been shown to cause oxidative damage in untrained persons while in trained persons such effects are not observed, possibly due to an increased resistance to oxidative stress [2,12]. It was suggested that the deleterious effects are reversible with proper rest [13]. Accordingly, strenuous physical exercise or prolonged intensive training, such as those performed by elite athletes, may result in overproduction of ROS, overwhelming the AD capacity, and consequently induce damage to macromolecules resulting in adverse effects on health and performance [7], such as contractile dysfunction that promotes muscle weakness and fatigue [14,15] while on the other hand regular training has positive effects to health by implementing cellular processes that lead to increased expression of antioxidants, hence inducing a mild oxidative stress that further stimulates the expression of certain antioxidant enzymes [16].

Biomarkers of oxidative stress are relevant in the evaluation of a disease status or health enhancing effects of antioxidants. Nevertheless, the scientific community has not reached a consensus regarding the best biomarkers to assess, quantify and diagnose ROS effects on the human body. It seems that a combined profile of several blood markers is desirable. The choice of the biomarkers should be dictated by the aim of the study and its design, as wells as by the clinical relevance for the selected population and their possible follow up, if applicable. Additionally, it should stain both a specific and an overall index of the redox status of the human body [17].

With the intent of studying ROS balance in men that undergo high volume and intensity exercises and in men that accomplish regular and moderate exercise, we aimed to investigate redox markers at rest of adult-competitive swimmers against recreational exercise practitioners. In the athletes at rest we expect to grasp if a cumulative effect of the periodic exposure to ROS exist or if the stimulus of the antioxidant defences can reverse the aforementioned cumulative effect.

2. Methods

2.1. Participants

Fifteen high competition male swimmers (S) and 16 active men (AM) participated in the study. They were all non-smokers and aged between 18—25 years.

All swimmers were members of sports clubs in the Lisbon area, had regular and systematic training, were included in the competition level for at least 5 years and found their weekly average hours of training in the interval between 17 and 23 h (21.5 ± 1.8 hours per week). In the days preceding the evaluation the training program was not changed. The members of the comparison group performed a maximum of 9 h of physical activity per week, due to their bachelor in sports science, but did not systematically practice any sport for at least one year.

All participants were asked about the use of antioxidant/anti-inflammatory drugs, nutraceuticals or enhance recovery drugs, with none of them taking any type of medication.

All participants were informed about the possible risks of the investigation before giving their written consent. All procedures were approved by the Ethic Committee of the Faculty of Human Kinetics, University of Lisbon and were conducted in accordance with the declaration of Helsinki for human studies [18].

2.2. Study design

A cross sectional profile, including anthropometry, body composition parameters, cardiorespiratory test and blood collection was performed in the early morning, at rest, followed by an overnight fast, in all participants. No alcohol or stimulant beverages were consumed in the previous day, and none of the subjects had participated in any exercise within the 12-hours prior to evaluation. Nutritional questionnaires were explained afterwards by a certified nutritionist.

2.3. Anthropometry

Weight and stature were assessed to the nearest 0.1 kg and 0.1 cm (Seca, Hamburg, Germany) according to standardized procedures [19]. Body mass index (BMI) was calculated as weight (kg) divided by the square of the stature (m).

2.4. Body composition

To characterize the sample, fat mass (FM) and fat-free mass (FFM) were determined by dual-energy X-ray absorptiometry (DXA) (Hologic, Waltham, USA), according to standardized procedures [20].

2.5. Cardiorespiratory fitness

A Maximal exercise (ME) trial was accomplished to evaluate cardiorespiratory performance of the participants.

A continuous graded maximal run was performed on a treadmill (Quinton Instrument Company, Seattle, Washington) with expired gas analysis (Cardiopulmonary Diagnostics Systems, Medical Graphics Corporation, Oak Grove Pkwy, Saint Paul, MN) and heart rate monitorization (Pacer ECG, POLAR, NY, USA) until O_2 uptake stabilization or voluntary exhaustion. At the ME trial, maximal oxygen uptake (VO_{2Max}), oxygen uptake at the anaerobic threshold (VO_{2AnaT}) and heart rate associated with VO_{2Max} (HR_{Max}) and with VO_{2AnaT} (HR_{AnaT}) were obtained. VO_2 was measured as an absolute value ($mL.min^{-1}$) and relative to weight ($mL.min^{-1}.kg^{-1}$).

2.6. Nutritional habits

Energy and nutrients intake were assessed from a 24-h diet record over a three-day period. A selection of some micronutrients was made according to their antioxidant potential (antioxidant vitamins and minerals acting as co-factor for antioxidant enzymes). Subjects were instructed regarding portion sizes, supplements, food preparation, and others aspects pertaining to an accurate recording of their intake. After the three-day period, all participants were interviewed by a qualified nutritionist in order to clarify the records. Diet records were analysed by the software package Food Processor SQL (Salem, Oregon, USA).

2.7. Biochemical analysis

Blood collection was performed in seated position from the antecubital vein, at rest, after an overnight fast, into tubes containing ethylenediamine-tetraacetic acid (EDTA) as anticoagulant.

Hemoglobin (Hb) in total blood was quantified by the method of Drabkin & Austin [21] and hematocrit (Hct) was determined by centrifugation (15250 × g for 7 min) in capillary tubes.

After, blood was centrifuged at 1000 g for 10 minutes, at 4 °C, for plasma and red blood cell (RBC) separation.

Plasma was used to asses oxidative damage markers: thiobarbituric acid reactive substances (TBARS) according to a modified method of Ohkawa et al. [22], activities of adrenaline oxidase (AdOx) according to Laires et al. [23], and of acid phosphatase (ACP1), according to Dissing [24]. Although the last two aforementioned parameters are not typical markers of oxidative damage, we have used these measures to complement TBARS. AdOx evaluates the ability of plasma to oxidise adrenaline, which may result from free ions or from leucocytes' activity, and ACP1 may contribute to the reduction of GR activity as it has a Flavin cofactor, such as GR, to which both enzymes may compete.

Additionally, plasma was used to assess muscle damage markers: creatine kinase (CK) as described in Laires et al. [23]; non-enzymatic antioxidants: α-tocopherol (VitE) according to procedures described by Lyan and Collegues [25] and total carotenoids (Carotenoids) according to the Rayssiguier group [26].

To assess RBC's susceptibility to peroxidation induced in vitro by H_2O_2 (RBC Px), cells were washed 3 times with NaCl 0.9% (w/v). RBC Px was evaluated by photometry with a method modified by the group of Rayssiguier [27]. The antioxidant enzymes activities were evaluated after hemolysis of RBC: superoxide dismutase (SOD) was determined according to Winterbourn et al. [28]; glutathione peroxidase (GPx) according to Paglia & Valentine [29], glutathione reductase (GR) according with Beutler [30], and meta-hemoglobin reductase (RmHb) according with Board [31].

2.8. Statistical analyses

Data were analysed with IBM SPSS Statistics for Windows version 24.0 (SPSS Inc, an IBM company, Chicago) and statistical significance was set at $P < 0.05$. Normality was verified using the Shapiro-Wilk test. Comparison between group means was performed using independent sample T-test and is presented as mean ± standard deviation. Bivariate correlations were performed to investigate the association between oxidative and muscle damage markers and enzymatic and non-enzymatic antioxidants. Partial correlation analyses were additionally conducted to control for the exercise level.

3. Results

Table 1 presents the physical characteristics and body composition variables for S an AM group. Swimmers showed higher FFM and lower FM, than AM.

Regarding biochemical parameters (Table 1), differences between groups were only observed for RBCPx ($P = 0.033$) and RmHb ($P = 0.033$) (lower in S).

Regarding nutritional habits (Table 2), our analysis revealed similar food intake between the two groups. Both groups presented mean values under the Reference Daily Allowance (RDA) for retinol and α-tocopherol and over the RDAs for iron, zinc, copper, and selenium; additionally AM presented low mean values for magnesium intake. The other selected micronutrients were in accordance with the RDA.

Results of the maximal exercise task are presented in Table 3. Expired gas analysis during the ME showed higher VO_{2Max} and VO_{2ANAT} for S, than for AM group.

Regarding redox balance when controlling for the exercise level (Table 4), AdOx was inversely correlated with Carotenoids and directly with CK and with TBARS; RBCPx was inversely correlated with ACP1 and directly correlated with GR; GR was inversely correlated with TBARS; and VitE was inversely correlated with TBARS and directly correlated with Carotenoids.

4. Discussion

In general, it has been reported that exercise induces an increase in antioxidant protein levels and in antioxidant activity [2]. In our study, mean values of all redox markers were similar between groups, with the exception of RBCPx and RmHb, which were lower in S, suggesting a lower susceptibility of erythrocytes to oxidation, in this group. These results suggest that the AD are effective at controlling ROS even if their production increases due to the exercise effect on-going in competitive swimmers. Additionally we explored if there was an influence of the nutritional habits, in particular regarding nutrients with antioxidant properties or

Table 1 Descriptive characteristics and body composition (mean ± SD).

	Competition swimmers (n = 15)	Active men (n = 16)
Age (years)	20.0 ± 1.7	21.1 ± 1.5
Weight (kg)	70.6 ± 5.3	71.1 ± 10.1
Height (cm)	176 ± 5.9	176 ± 7.5
BMI (kg/m^2)	22.7 ± 1.6	23.0 ± 3.0
FM (%)*	10.4 ± 3.3	17.5 ± 6.9
FFM (kg)*	62.6 ± 4.4	58.2 ± 5.7
Hb (g/dL)	16.2 ± 1.2	15.9 ± 0.9
Hct (%)	47.8 ± 3.0	46.8 ±1.9
TBARS (μmol/dm^3)	3.18 ± 0.94	3.20 ± 1.00
AdOx (μmol/2 h.L)	23.6 ± 13.4	28.1 ± 20.0
ACP1 (μmol/g Hb/h)	206.9 ± 57.9	162.6 ± 31.1
RBCPx (%)*	39.2 ± 4.8	46.3 ± 9.5
CK (U/L)	144.0 ± 127.0	86.0 ± 55.4
RmHb (μmol/min.g Hb)*	7.5 ±2.1	9.1 ± 2.3
GR (mol/min.g Hb)	89.7 ± 14.1	86.6 ± 13.9
GPx (mol/min.g Hb)	103.0 ± 24.2	115.0 ± 25.2
SOD (U/g Hb)	2222 ± 411	2198 ± 410
VitE (μmol/dm^3)	22.4 ± 6.07	23.2 ± 6.35
Carotenoids (nmol/dm^3)	644 ± 375	898 ± 651

BMI: body mass index; FM: fat mass; FFM: fat-free mass; Hb, hemoglobin; Hct, hematocrit; TBARS: thiobarbituric acid reactive substances; AdOx: adrenaline oxidase activity; ACP1, acid phosphatase; Hb, Hemoglobin; RBCPx: susceptibility of red blood cells to in vitro peroxidation; CK: creatine kinase; RmHb: meta-hemoglobin reductase; GR: glutathione reductase; GPx: glutathione peroxidase; SOD: superoxide dismutase; VitE: alpha-tocopherol; Carotenoids: total carotenoids.

* Significant difference between groups (P < 0.05).

Table 3 Cardiorespiratory variables (mean ± SD) obtained in the maximal task (ME).

	Competition swimmers (n = 15)	Active men (n = 16)
VO_{2Max} (mL.min^{-1}.kg^{-1})*	51.3 ± 7.1	43.2 ± 5.7
HR_{Max} (beat.min^{-1})	190 ± 8	194 ± 6
VO_{2AnaT} (mL.min^{-1}.kg^{-1})*	38.0 ± 4.4	29.2 ± 5.1
HR_{AnaT} (beat.min^{-1})	162 ± 10	157 ± 15
VO_{2AnaT}/VO_{2Max} (%)*	74.6 ± 7.1	67.4 ± 5.6

VO_2: oxygen volume; HR: heart rate.

* Significantly different between groups (P < 0.05).

Table 4 Partial correlations between redox balance parameters, controlling for exercise level.

	Controlling for exercise level
AdOx	
Carotenoids	R = −0.596 (P = 0.001)
CK	R = 0.413 (P = 0.045)
TBARS	R = 0.532 (P = 0.005)
RBCPx	
ACP1	R = −0.392 (P = 0.048)
GR	R = 0.524 (P = 0.006)
GR	
TBARS	R = −0.404 (P = 0.041)
Vit E	
TBARS	R = −0.543 (P = 0.004)
Carotenoids	R = 0.723 (P < 0.001)

AdOx: adrenaline oxidase activity; CK: creatine kinase; TBARS: thiobarbituric acid reactive substances; carotenoids: total carotenoids; RBCPx: susceptibility of red blood cells to in vitro peroxidation; ACP1: acid phosphatase; GR: glutathione reductase; VitE: alpha-tocopherol.

Table 2 Nutritional habits (mean ± SD).

Variable	Competition Swimmers (n = 15)	Active men (n = 16)	Reference daily allowance [51–56]
Calories (Kcal.day^{-1})	2976 ± 744	2845 ± 450	–
Calories (Kcal.day^{-1}.kg^{-1})	42.8 ± 10.2	40.6 ± 7.34	–
Linoleic Acid/Linolenic acid ratio	8.83 ± 2.22	10.3 ± 2.67	5/1
Water (mL.day^{-1})	2337 ± 534	2078 ± 740	–
Magnesium (mg.day^{-1})	342 ± 164	319 ± 62.3[a]	410
Iron[a] (mg.day^{-1})	24.3 ± 9.69	20.7 ± 6.93	11
Zinc[a] (mg.day^{-1})	14.5 ± 4.69	13.5 ± 3.93	11
Copper[a] (μg.day^{-1})	1336 ± 499	1255 ± 339	890
Selenium[a] (μg.day^{-1})	101 ± 48.0	102 ± 39.1	55
Manganesium (mg.day^{-1})	2.61 ± 2.65	2.71 ± 2.14	2.2
Retinol[a] (μg.day^{-1})	433 ± 289	412 ± 199	900
Vitamin C (mg.day^{-1})	136 ± 120	108 ± 70.4	75
α-tocopherol[a] (mg.day^{-1})	5.75 ± 3.00	6.02 ± 2.21	15

[a] Significantly different from reference daily allowance.

known to affect the redox balance. Also anti-inflammatory drugs were disclosed to ensure that redox balance was not compromised by external factors, other than training. Both groups had similar nutritional patterns and no participant was taking external medication. The low number of participants of the present study is a potential limitation. However, studies with athletes of one sport hardly recruit large samples of participants, especially if they involve invasive data collection techniques such as blood sampling.

Physiological outcomes regarding oxygen kinetics indicated, as expected, differences between groups regarding the response to the maximal task (with higher VO_{2Max} and VO_{2ANAT} in swimmers). The higher values of maximal oxygen consumption relate to the higher capacity of athletes to integrate and optimize oxygen capture, fixation, transport and utilization capacity.

In order to ensure that swimmers did not have technical advantage over controls in the cardiorespiratory fitness test, and bearing in mind that a single bout of exercise has a limited adaptive response with the increased ROS level being the important stimulus for muscle cells to adapt to chronic exercise [32], the maximal exercise task was performed in a treadmill. The treadmill is a very unspecific endurance test for swimmers and its results might consequently underestimate the aerobic performance of the swimmers, nevertheless this test is the one that best relates to the muscular patterns of day-to-day movements. Additionally, the treadmill involves almost all muscular groups in the task, this way representing a good estimation of the cardiorespiratory capacity. Considering the group differences for RBCPx and RmHb, our data suggest a greater renewal of RBCs in the swimmers' group, which may be advantageous in order to prevent the decline of the viable RBC number [33] as swimmers presented a significantly lower mean value for RBCPx, (with a power of 0.727 and an effect size of 0.943 — large) and a significantly lower mean value for the activity of RmHb, which increases when methemoglobin increases as a result of oxidative stress (with a power of 0.497 and an effect size of 0.730 — medium). Thus the lower value of RmHb activity in swimmers may also point out to a smaller production of meta-hemoglobin in the RBC. Meta-hemoglobin formation is accompanied by formation of superoxide anion. The aforesaid anion has relatively low reactivity but can originate other radicals with higher reactivity, therefore increasing RBCs damage. Moreover, the accumulation of meta-hemoglobin in RBCs may lead to the formation of Heinz bodies which, in turn, can render erythrocytes less flexible and deformable, thus conditioning their average life time by premature reticulo-endothelial entrapment, especially at the spleen level [34], hence increasing the probability of intravascular hemolysis occurrence [33,35,36]. This suggests that a higher production of meta-hemoglobin may be associated with a higher fragility of the RBCs. According to Smith [33], in athletes it is frequently observed an oxidant pressure, along with osmotic and acid-base balance alterations. These physiological changes may lead to a decrease in the average lifetime of erythrocytes and an increase in erythropoiesis rate. The above-mentioned author observed an increase in reticulocyte numbers two days after a marathon, followed by a decrease of plasma haptoglobin levels [37]. A higher percentage of young RBC may favour athletes as these cells are more efficient in oxygen transportation [38]. In younger erythrocytes, hemoglobin is probably more stable, therefore resulting in a lower amount of meta-hemoglobin and superoxide anion radical production. At this regard, RmHb may be less required and RBC may be less susceptible to in vitro peroxidation induced by H_2O_2. Thus, we could risk suggesting that swimmers are protected against oxidative stress, if it weren't for the antioxidant enzymes activities that were not significantly different.

In unfatigued muscle, intracellular ROS appear to be essential for, and even seem to increase, normal force generation [2]. Some uncertainty surrounds the effect of ROS on the force generating myofilaments. Initial data have shown that brief exposure to low concentrations of H_2O_2 increased force by 27%, while a longer exposure results in force decline [39]. Even though the exact mechanisms are unidentified, it is generally assumed that this may result from oxidative modifications of a variety of proteins in intracellular components and Ca^{2+} release channels. At this concern, it was suggested that troponin I (which is involved in sensing intracellular Ca^{2+} levels) is possibly a target of ROS. Nonetheless, ROS may also lead to changes in contractile proteins, namely myosin [2,14,15].

Besides their effects on the contractile kinetics, ROS are also able to modulate various signalling pathways, such as calcium, protein tyrosine kinases and phosphatases, serine/threonine kinases, and phospholipases [2]. Evidence suggest that exercise-induced ROS are primarily of non-mitochondrial origin and involve several physiological processes of ROS generation and propagation, particularly nicotinamide adenine dinucleotide phosphate (NAPH) oxidases [11,40]. Enhancement of ROS induced damage, in particular lipoperoxidation, can also be due to the increased release of catecholamines, because during their oxidation to adrenochrome, oxygen free radicals are formed, which in turn induce oxidative damage/lipoperoxidation [23,41]. Adrenochrome may also inhibit the electron transport chain in mitochondria, by interacting with quinones and reducing respiratory control, and induce the release of calcium from endoplasmic reticulum, [42]. This theory may explain the direct correlation between AdOx and CK as the leakage of this enzyme to the plasma is often explained by mechanisms involving increased cytoplasm calcium concentration [23,42,43]. Additionally, this correlation may be a result of the pro-oxidant character of AdOx and reinforce AdOx's utility as an oxidative damage marker [41]. Bearing in mind that: our results were obtained including a control group; AdOx was not statistically different between groups but a tendency to be smaller in swimmers was observed; and that swimmers have an exercise practice with a low mechanical impact, we may suggest that AdOx may have a relevant role in reducing the activation of oxidative damage mechanisms such as intravascular hemolysis, metal ions leakage into plasma, or even the inflammatory response [44—47]. AdOx was also directly associated with TBARS and inversely with Carotenoids. These additional associations reinforce and suggest that, in both groups, plasma Carotenoids play a primordial and essential role in the circulating AD, once

the carotenoids concentration in plasma is inversely proportional to the plasmatic oxidative damage markers (AdOx and TBARS). Additionally, TBARS were inversely associated with VitE. Bearing in mind that VitE is lipossoluble and that TBARS are fundamentally lipidic oxidative damage markers, this association reinforce the need for an active lipossoluble non-enzymatic antioxidant system. The direct association of Carotenoids with VitE probably due to the lipossoluble nature of both substances support the role VitE and Carotenoids as a primary line of defence against ROS. The aforementioned correlation between oxidative damage markers (TBARS) and non-enzymatic antioxidants (VitE) seem to highlight the importance of these AD in plasma [2].

Nevertheless, we would like to reinforce that although TBARS is a simple assay for lipid oxidation quantification most frequently used, the pink coloured products formed as biological samples react with Thiobarbituric Acid that is posteriorly measured by colorimetry or fluorometry cannot be interpreted as a diagnostic index of the extent of solely lipid peroxidation, fatty hydroperoxide formation, or oxidative injury to tissue lipids, as other molecules also react with thiobarbituric acid. However, despite TBARS being a poor marker of lipid peroxidation by itself, when combined with other oxidative damage markers, muscle damage markers, non-enzymatic antioxidants and antioxidant enzymes, can serve as a prognostic marker for redox balance [48,49], and give, when combined, an overall index of the redox status in a particular condition [17].

The direct association between GR and RBCPx observed, and the inverse association observed between GR and TBARS, led us to speculate that higher GR activity resulting from an increase in the oxidant status may be accompanied by an increase of RBC's damage markers.

As ACP1 has a flavin mononucleotide phosphatase activity, it may contribute to lower the bioavailability of flavin adenine dinucleotide, the cofactor for GR. As this enzyme is responsible for recycling GSH [50], the lowering of this substance may contribute to render RBC more susceptible to ROS. The inverse association between RBCPx and ACP1 corroborate this principle. The paradoxical situation that increased exercise-induced oxidative stress causes beneficial effects, possibly by providing an local adaptation to the detrimental agent, and consequently is responsible for systemic beneficial effects, including an improvement of the general physiological function, a reduction in disease prevalence, and a higher quality of life, has been recently a focus of discussion [2]. The beneficial effects appear to depend upon the duration of the exercise, leading to a broad conclusion that a single bout of exercise is suggested to lead to a limited adaptive response, and regular exercise appears to gradually increase the level of adaptation by the repeated activation of antioxidant genes and proteins [32]. The aforementioned authors [32] theorize that it is the increased ROS level that is the important stimulus for muscle cells to adapt to chronic exercise. The improved capacity to decrease ROS may then provide a better protection during subsequent training bouts and also temper the aging process and promote healthy functional capacities [32]. Notwithstanding, uncontrolled production of ROS can damage cells and eventually be a liability to health. Intracellular oxidants also play regulatory roles in the modulation of skeletal muscle force production, regulation of cell signalling pathways and are central for gene expression. As a consequence, exacerbated AD may engulf the need for ROS elimination therefore reducing their benefit effects, which may also be detrimental to health.

5. Conclusions

In conclusion, our results reinforce the importance of VitE and Carotenoids as protective lipid antioxidants both in athletes and in active men performing recreational sports activities, and suggest that increased oxidative imbalance, resulting in higher TBARS, may be associated to the stimulation of GR activity, probably as a result of oxidized glutathione (GSSG) increase. In addition, molecular muscle leakage, to which oxidative damage contributes, is associated to the AdOx activity of plasma, which may result from ion leakage, intravascular hemolysis or inflammation. Therefore, our results suggest that competitive swimmers are not at higher risk of oxidative damage than recreational exercise practitioners.

Disclosure of interest

The authors declare that they have no competing interest.

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